

Section I**10 Marks****Use the multiple-choice answer sheet for Questions 1 - 10**

1) $\frac{8.964}{\sqrt{61.328}}$ expressed to 3 significant figures is:

- A** 1.145 **B** 1.14 **C** 1.144 **D** 7.83

2) $p^3 - 8$ factorises to give:

- A** $(p - 2)^3$ **B** $(p - 2)(p^2 + 2p + 4)$
- C** $(p - 2)(p^2 - 2p + 4)$ **D** $(p - 2)(p^2 + 4p + 4)$

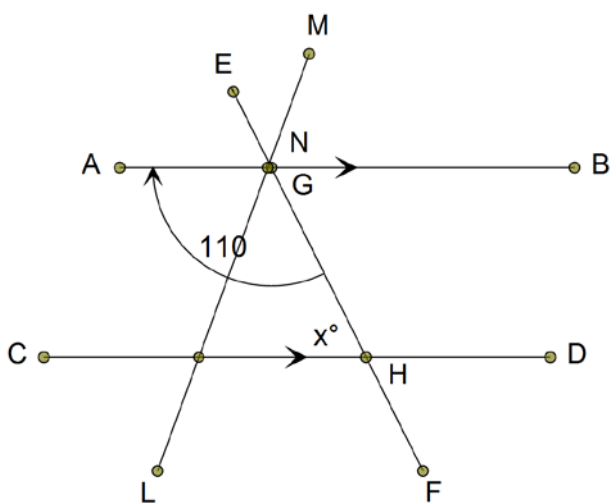
3) If $f(x) = 5 - 3x - 2x^2$ then $f(-2)$ is equal to:

- A** 3 **B** -11 **C** 5 **D** -7

4) The largest possible domain of $y = \sqrt{4 - x^2}$ is

- A** All real numbers **B** $x \leq 2$
- C** $-2 \leq x \leq 2$ **D** $x > -2$ and $x < 2$

5) The value of x is



A 60°

B 70°

C 50°

D 65°

6) Which of the following has solutions 4 and -3 ?

A $x^2 + x - 12 = 0$

B $x^2 - x - 12 = 0$

C $x^2 - 7x + 12 = 0$

D $x^2 + 7x - 12 = 0$

7) The solution for $x^2 - 49 \geq 0$ is

A $x \leq -7$ and $x \geq 7$

B $x \geq -7$ and $x \geq 7$

C $x \leq -7$ and $x \leq 7$

D $-7 \leq x \leq 7$

8) $(2\sqrt{5} - 1)^2$ is equal to

A 19

B $11 - 2\sqrt{5}$

C $19 - 2\sqrt{5}$

D $21 - 4\sqrt{5}$

9) The perpendicular distance from the line $3x - 4y + 6 = 0$ to the point $(-1, -3)$ is

A $\frac{3}{5}$

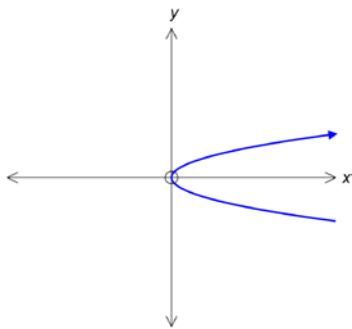
B 3

C $\frac{4}{3}$

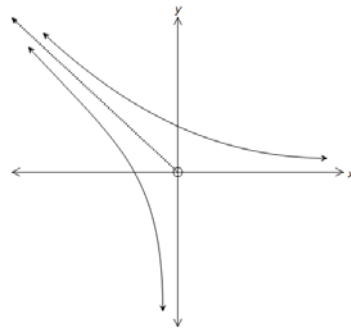
D 15

10) Which of the following graphs represent a function ?

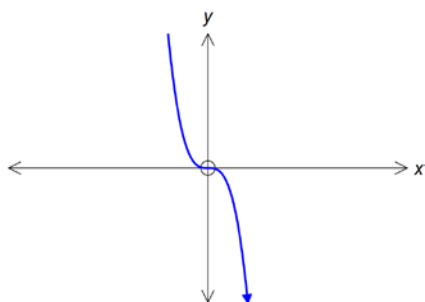
A



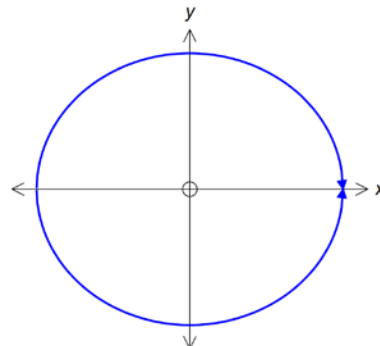
B



C



D



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Section II**Attempt Questions 11 to 16**

Answer each question in separate writing booklets.

All necessary working should be shown in every question.

Question 11 (14 Marks)	Marks
a) Express $0.02\dot{4}$ as a fraction in its simplest form.	2
b) Solve simultaneously $3x - 2y = 29$ $4x + 5y = 8$	2
c) Solve $ 3x - 8 < 1$	2
d) Solve $\frac{x+1}{3} + \frac{x}{4} = 5$	2
e) Simplify $\frac{x^2 - 144}{x^2 + 15x + 36} \div \frac{2x^2 - 24x}{(x+3)^2}$	3
f) Find the exact solution(s) to $4x^2 + 12x + 1 = 0$ in simplest form.	3

Question 12 (13 Marks)Start a new booklet.**Marks**

a) Sketch the following showing all essential features

i) $x^2 + y^2 = 9$ **2**

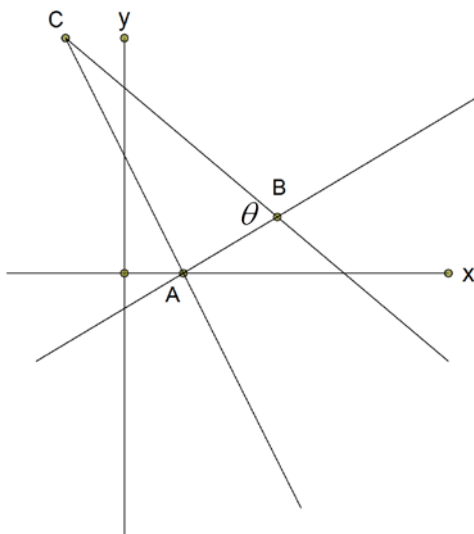
ii) $y = \frac{1}{x+3}$ **2**

iii) $y = |x| + 3$ **2**

iv) $y = 3^{-x}$ **2**

b) Sketch the parabola $y = x^2 - 3x - 18$. Clearly showing the coordinates of the x and y intercepts and the vertex. **3**

c) Show that $f(x) = x^5 - 16x$ is an odd function. **2**

Question 13*(13 Marks)*Start a new booklet.**Marks**

The diagram shows the points $A(1,0)$ $B(4,1)$ and $C(-1, 6)$ in the Cartesian plane. Angle ABC is θ .

Copy and trace the diagram into your writing booklet.

- | | |
|---|----------|
| a) Show that C lies on the line $3x + y = 3$. | 1 |
| b) Find the gradient of AB . | 1 |
| c) Find the equation of AB . | 1 |
| d) Find the length of AB . | 1 |
| e) Show that AB and AC are perpendicular. | 2 |
| f) Find the area of $\triangle ABC$. | 2 |
| g) Find $\tan \theta$. | 1 |
| h) Find the equation of the circle with centre A the passes through B . | 1 |
| i) The point D is not shown on the diagram. The point D lies on the line $3x + y = 3$ between A and C , $AD = AB$. Find the coordinates of D . | 2 |
| j) On your diagram shade the region satisfying the inequality $3x + y \leq 3$. | 1 |

Question 14 (17 Marks)Start a new booklet. **Marks**

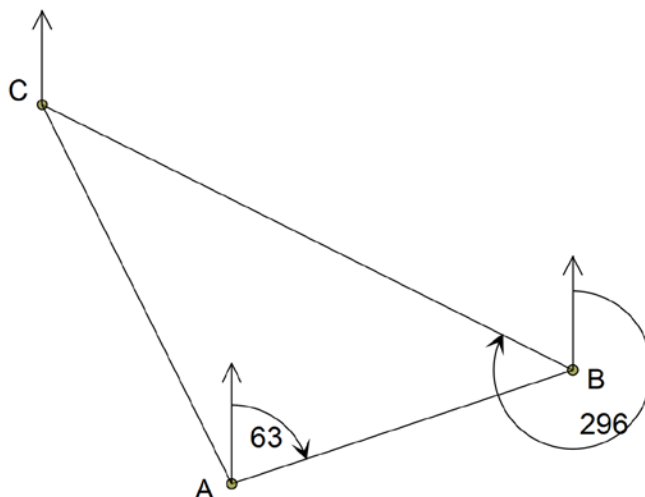
a) Find the exact value of $\cos 150^\circ$ **1**

b) Solve each equation for $0^\circ \leq \theta \leq 360^\circ$

i) $\cos^2 \theta = \frac{1}{2}$ **2**

ii) $\tan 2x = \sqrt{3}$ **2**

c) A ship sails 50km from port A to port B on a bearing of 063° T then sails 130km from port B to port C on a bearing of 296° T.



i) Explain why $\angle ABC = 53^\circ$. **2**

ii) Find, correct to the nearest km, the distance of port A from port C. **2**

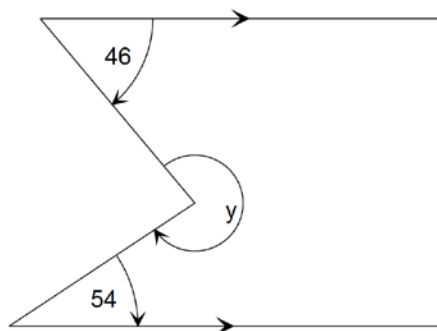
iii) Find the bearing of port A from port C. Answer correct to the nearest minute. **3**

d) Prove that $\tan \theta + \cot \theta = \sec \theta \csc \theta$. **3**

e) If $\cos \alpha = \frac{4}{5}$ and $\sin \alpha < 0$, find $\tan \alpha$. **2**

Question 15 (12 Marks)Start a new booklet.**Marks**

- a) Find the value of
- y
- , giving reasons.

2

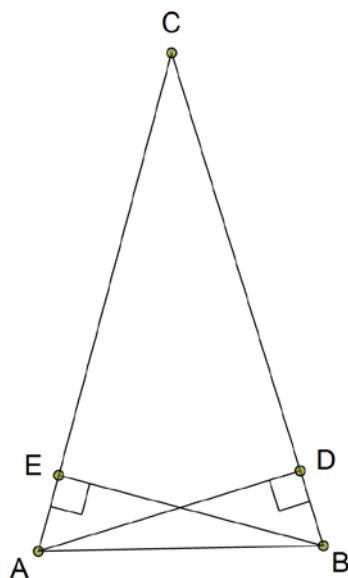
- b) Find the size of each interior angle of a regular octagon.

2

- c) Let
- AD
- and
- BE
- be the two altitudes of triangle
- ABC
- . Suppose that
- $AD=BE$

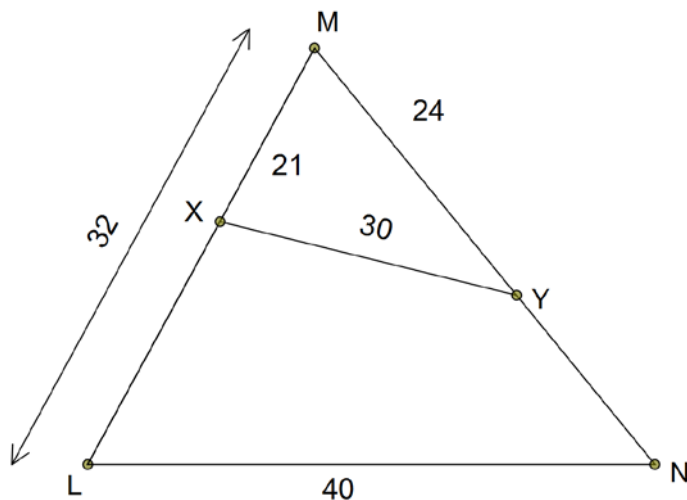
4

- i) Prove that $\triangle ABE \cong \triangle BAD$.
- ii) Hence prove that $\triangle ABC$ is isosceles.



Question 15 continued.**Marks**

- d) i) Prove that $\triangle LMN$ and $\triangle YMX$ are similar.
 ii) Hence find YN.

4**Question 16** (21 Marks)[Start a new booklet.](#)

- a) Differentiate the following with respect to x

i) $y = 5x^4 - 7x^2 + 8$ **1**

ii) $y = \frac{3x^4 - 5x^2}{x}$ **2**

iii) $y = \frac{1}{x\sqrt{x}}$ **2**

v) $y = \frac{6}{x^4}$ **1**

vi) $y = (3x + 4)^5$ **2**

vii) $y = x(3 - 2x)^4$ **2**

viii) $y = \frac{x^2 + 5}{x - 2}$ **2**

ix) $y = x\sqrt{1 - x^2}$, answer without fractional or negative indices **3**

Question 16 Continued.**Marks**

- b) A function $f(x) = x^2 + 4x - 12$ has a tangent with a gradient of -6 at the point P on the curve. Find the coordinates of P. **3**
- c) Find the equation of the normal to the curve $y = x^2 - 8x + 7$ at the point $(3, -8)$. **3**

End of paper

Section 1

$$1. \frac{8.964}{\sqrt{61.328}} \div 1.1446493$$

$$= 1.14 \text{ to 3 sig fig } \textcircled{B}$$

$$2. p^3 - 8 = p^3 - 2^3$$

$$= (p-2)(p^2+2p+4) \textcircled{B}$$

$$3. f(x) = 5 - 3x - 2x^2$$

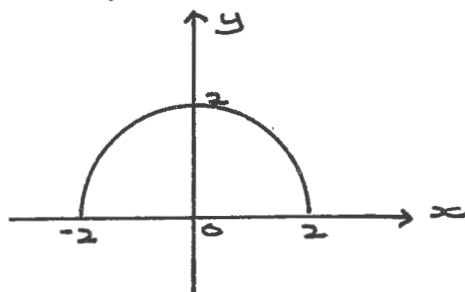
$$f(-2) = 5 - 3(-2) - 2(-2)^2$$

$$= 5 + 6 - 8$$

$$= 3$$

$\textcircled{A}$

$$4. y = \sqrt{4-x^2}$$



$$\therefore \text{Domain } -2 \leq x \leq 2 \textcircled{C}$$

$$5. x = 70$$

co-interior angles add
up to 180° $AB \parallel CD$

$\textcircled{B}$

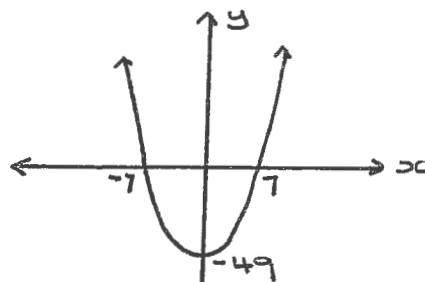
$$6. x = 4, -3$$

$$\therefore (x-4)(x+3) = 0$$

$$x^2 - x - 12 = 0$$

$\textcircled{B}$

$$7. x^2 - 49 \geq 0$$



$$\therefore x \leq -7, x \geq 7 \textcircled{A}$$

$$8. (2\sqrt{5}-1)^2 = (2\sqrt{5}-1)(2\sqrt{5}-1)$$

$$= 4 \times 5 - 2\sqrt{5} - 2\sqrt{5} + 1$$

$$= 21 - 4\sqrt{5} \textcircled{D}$$

$$9. 3x - 4y + 6 = 0 \quad (-1, -3)$$

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|3x - 1 - 4x - 3 + 6|}{\sqrt{3^2 + 4^2}}$$

$$= \frac{|-3 + 12 + 6|}{5}$$

$$= \frac{15}{5}$$

$$= 3$$

$\textcircled{B}$

10 A graph represents a function if a vertical line cuts the graph once

$\textcircled{C}$

Section 2

Question 11

a) let $x = 0.024$

$$= 0.024444\ldots$$

$$1000x = 24.444\ldots$$

$$100x = 2.444\ldots$$

$$900x = 22$$

$$x = \frac{22}{900}$$

$$= \frac{11}{450}$$

$$\therefore 0.024 = \frac{11}{450}$$

b) $3x - 2y = 29$ — ①

$$4x + 5y = 8$$
 — ②

$$\textcircled{1} \times 4 \quad 12x - 8y = 116$$
 — ③

$$\textcircled{2} \times 3 \quad 12x + 15y = 24$$
 — ④

$$\textcircled{3} - \textcircled{4} \quad -23y = 92$$

$$y = -4$$

$$3x - 2y = 29$$

$$3x - 2(-4) = 29$$

$$3x = 21$$

$$x = 7$$

$$\therefore x = 7, y = -4$$

c) $|3x - 8| < 1$

$$-1 < 3x - 8 < 1$$

$$7 < 3x < 9$$

$$\frac{7}{3} < x < 3$$



d) $\frac{x+1}{3} + \frac{x}{4} = 5$

$$4(x+1) + 3x = 60$$

$$4x + 4 + 3x = 60$$

$$7x = 56$$

$$x = 8$$

e) $\frac{x^2 - 144}{x^2 + 15x + 36} \div \frac{2x^2 - 24x}{(x+3)^2}$

$$= \frac{(x-12)(x+12)}{(x+12)(x+3)} \times \frac{(x+3)(x+3)}{2x(x-12)}$$

$$= \frac{x+3}{2x}$$

f) $4x^2 + 12x + 1 = 0$

$$a = 4, b = 12, c = 1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-12 \pm \sqrt{12^2 - 4 \times 4 \times 1}}{2(4)}$$

$$= \frac{-12 \pm \sqrt{128}}{8}$$

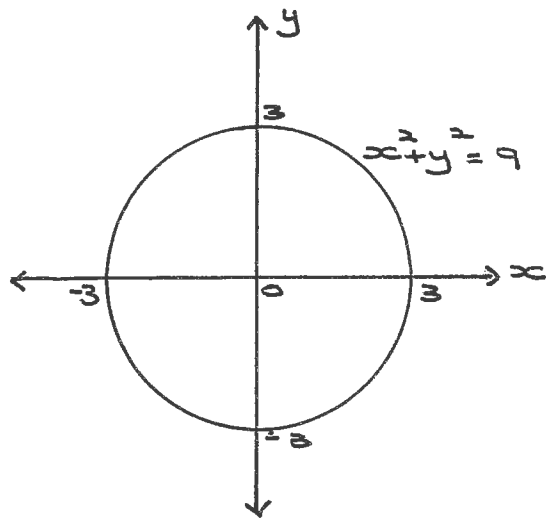
$$= \frac{-12 \pm 8\sqrt{2}}{8}$$

$$= \frac{4(-3 \pm 2\sqrt{2})}{8}$$

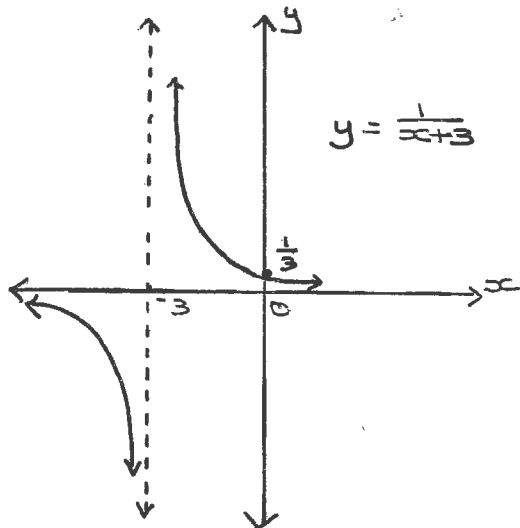
$$= \frac{-3 \pm 2\sqrt{2}}{2}$$

Question 12

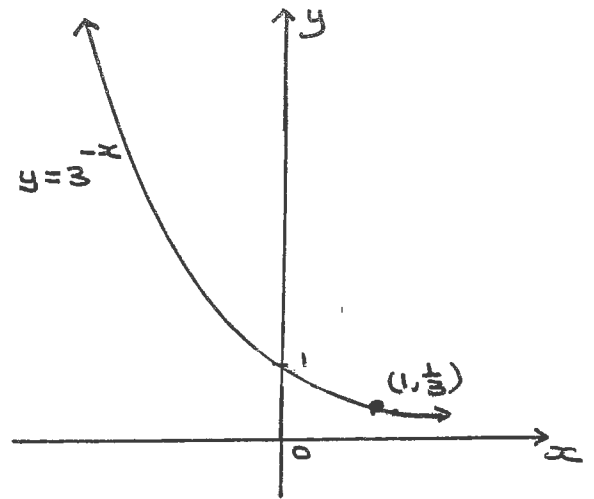
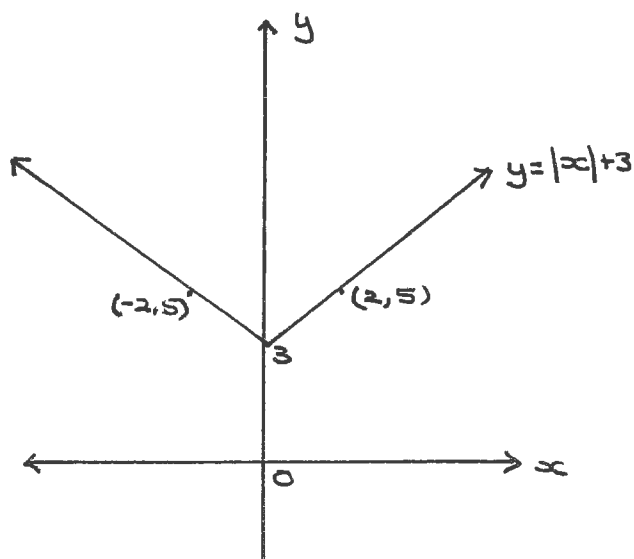
a) i)



ii)



iii)



b) $y = x^2 - 3x - 18$

x-intercepts $\Rightarrow y = 0$

$$x^2 - 3x - 18 = 0$$

$$(x-6)(x+3) = 0$$

$$x = 6, -3$$

$\therefore$ x intercepts are (6, 0) (-3, 0)

y-intercept $\Rightarrow x = 0$

$$y = x^2 - 3x - 18$$

$$= -18$$

$\therefore$ y-intercept is (0, -18)

equation of the axis of symmetry

$$x = \frac{-b}{2a}$$

$$x = \frac{-(-3)}{2}$$

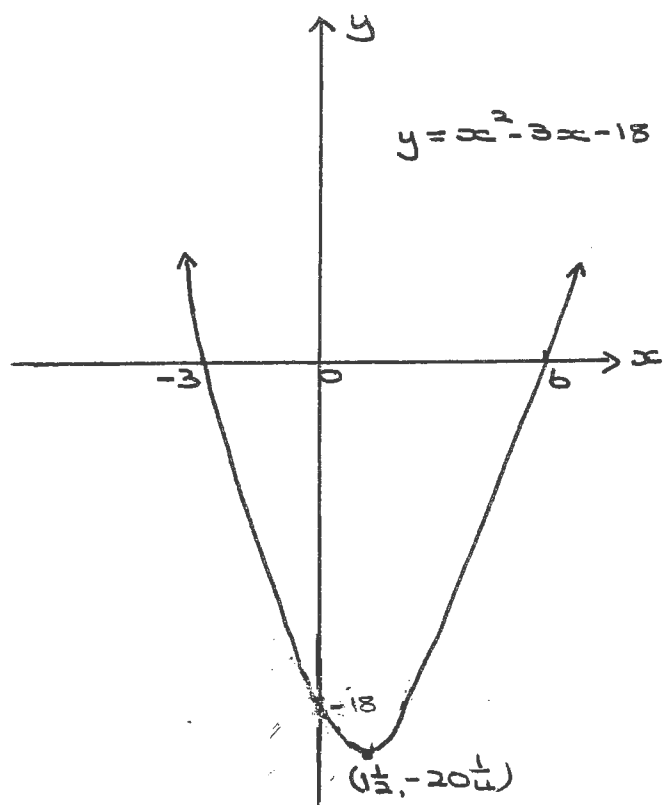
$$x = \frac{3}{2}$$

$$y = x^2 - 3x - 18 \text{ when } x = \frac{3}{2}$$

$$= \left(\frac{3}{2}\right)^2 - 3\left(\frac{3}{2}\right) - 18$$

$$= -20\frac{1}{4}$$

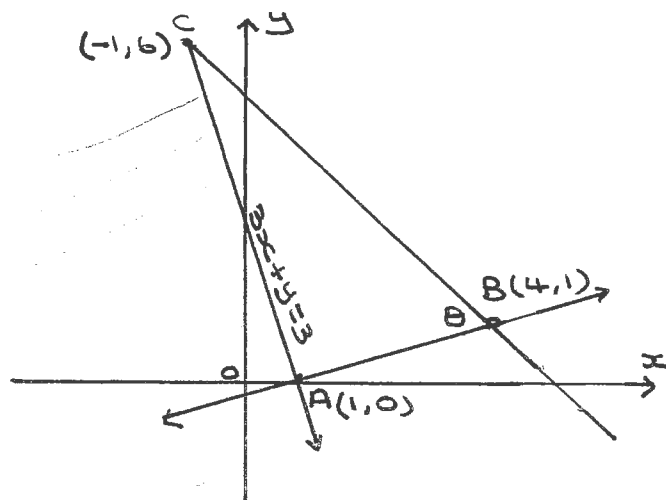
$\therefore$ vertex is $\left(1\frac{1}{2}, -20\frac{1}{4}\right)$



c) $f(x) = x^5 - 16x$
 $f(-x) = (-x)^5 - 16(-x)$
 $= -x^5 + 16x$
 $-f(-x) = -(-x^5 + 16x)$
 $= x^5 - 16x$

As $f(x) = -f(-x)$ the function is odd.

Question 13



a) $3x + y = 3$ C is $(-1, 6)$

$$\begin{aligned} \text{LHS} &= 3x + y \\ &= 3(-1) + 6 \\ &= 3 \\ &= \text{RHS} \end{aligned}$$

$\therefore$ C lies on the line $3x + y = 3$

b) $m_{AB} = \frac{y_2 - y_1}{x_2 - x_1}$ A(1,0) B(4,1)

$$= \frac{1 - 0}{4 - 1}$$

$$= \frac{1}{3}$$

$\therefore$ gradient AB is $\frac{1}{3}$

c) $y - y_1 = m(x - x_1)$ $m = \frac{1}{3}$ A(1,0)

$$y - 0 = \frac{1}{3}(x - 1)$$

$$y = \frac{1}{3}x - \frac{1}{3}$$

d) $d_{AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$= \sqrt{(4 - 1)^2 + (1 - 0)^2}$$

$$= \sqrt{10}$$

$\therefore$ length AB is $\sqrt{10}$ units

e) $m_{AB} = \frac{1}{3}$ from part (b)

$$m_{AC} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{6 - 0}{-1 - 1}$$

$$= -3$$

$$m_{AB} \times m_{AC} = \frac{1}{3} \times -3$$

$$= -1$$

$\therefore AB \perp AC$

$$\begin{aligned}
 f) d_{AC} &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\
 &= \sqrt{(-1 - 1)^2 + (6 - 0)^2} \\
 &= \sqrt{40} \\
 &= 2\sqrt{10}
 \end{aligned}$$

$$\begin{aligned}
 A &= \frac{1}{2}bh \\
 &= \frac{1}{2} \times AC \times AB \\
 &= \frac{1}{2} \times 2\sqrt{10} \times \sqrt{10} \\
 &= 10 \\
 \therefore \text{Area } \triangle ABC &\text{ is } 10 \text{ units}^2
 \end{aligned}$$

$$\begin{aligned}
 g) \tan \theta &= \frac{AC}{AB} \\
 &= \frac{2\sqrt{10}}{\sqrt{10}} \\
 &= 2
 \end{aligned}$$

$$\begin{aligned}
 h) (x-h)^2 + (y-k)^2 &= r^2 \\
 \text{centre } A(1,0) \quad r &= \sqrt{10} \\
 \therefore (x-1)^2 + y^2 &= 10
 \end{aligned}$$

$$\begin{aligned}
 i) \text{Length } AD &= \text{length } AB \\
 &= \sqrt{10} \\
 \text{length } AC &= 2\sqrt{10} \\
 \therefore D &\text{ is the midpoint of } AC \\
 D &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \\
 &= \left(\frac{1 + (-1)}{2}, \frac{0 + 6}{2} \right) \\
 &= (0, 3)
 \end{aligned}$$

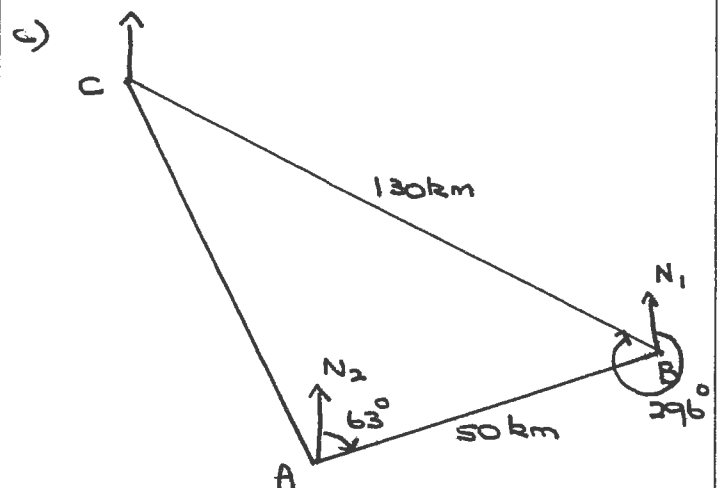
$$\begin{aligned}
 j) \text{ test } (0,0) &\text{ in } 3x + y \leq 3 \\
 &\quad 0 \leq 3 \text{ true} \\
 \therefore &\text{ shade side of line} \\
 &\text{ containing } (0,0) \\
 &\text{ see previous page}
 \end{aligned}$$

Question 14

$$\begin{aligned}
 a) \cos 150^\circ &= -\cos 30^\circ \\
 &= -\frac{\sqrt{3}}{2}
 \end{aligned}$$

$$\begin{aligned}
 b) i) \cos^2 \theta &= \frac{1}{2} \\
 \cos \theta &= \pm \frac{1}{\sqrt{2}} \\
 \text{Related angle } \cos 45^\circ &= \frac{1}{\sqrt{2}} \\
 \therefore \theta &= 45^\circ, 180^\circ - 45^\circ, 180^\circ + 45^\circ, 360^\circ - 45^\circ \\
 \theta &= 45^\circ, 135^\circ, 225^\circ, 315^\circ
 \end{aligned}$$

$$\begin{aligned}
 ii) \tan 2\theta &= \sqrt{3} \quad 0 \leq \theta \leq 360 \\
 &\quad 0 \leq 2\theta \leq 720 \\
 \text{Related angle } \tan 60^\circ &= \sqrt{3} \\
 \therefore 2\theta &= 60^\circ, 180^\circ + 60^\circ, 360^\circ + 60^\circ, 540^\circ + 60^\circ \\
 &\quad 360^\circ + 180^\circ + 60^\circ \\
 2\theta &= 60^\circ, 240^\circ, 420^\circ, 600^\circ \\
 \theta &= 30^\circ, 120^\circ, 210^\circ, 300^\circ
 \end{aligned}$$



$$N_1BC = 64^\circ \text{ angles about a point add up to } 360^\circ$$

$$N_1BA = 117^\circ \text{ co-interior angles add up to } 180^\circ \text{ } N_1B \parallel N_2A$$

$$\therefore \angle ABC = 53^\circ \text{ by subtraction}$$

$$\begin{aligned}
 \text{ii) } b^2 &= a^2 + c^2 - 2ac \cos B \\
 &= 130^2 + 50^2 - 2 \times 130 \times 50 \cos 53^\circ \\
 &\therefore 11576.4047 \\
 b &\therefore 107.5937019
 \end{aligned}$$

$\therefore$ distance is 108 km to nearest km

$$\text{iii) } \frac{\sin C}{c} = \frac{\sin B}{b}$$

$$\frac{\sin C}{50} = \frac{\sin 53}{b}$$

$$\sin C = \frac{50 \sin 53}{b}$$

$$C = 21^\circ 47' \text{ to nearest min}$$

$$\begin{aligned}
 \therefore \text{Bearing is } (116 + 21^\circ 47')^\circ \text{T} \\
 = 137^\circ 41' \text{T}
 \end{aligned}$$

$$\text{d) } \tan \theta + \cot \theta = \sec \theta \operatorname{cosec} \theta$$

$$\text{L.H.S} = \tan \theta + \cot \theta$$

$$= \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta}$$

$$= \frac{1}{\cos \theta \sin \theta}$$

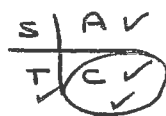
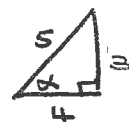
$$= \sec \theta \operatorname{cosec} \theta$$

$$= \text{R.H.S}$$

$$\therefore \tan \theta + \cot \theta = \sec \theta \operatorname{cosec} \theta$$

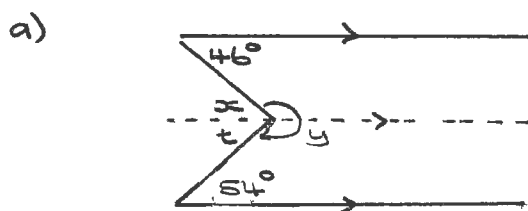
$$\text{2) } \cos \alpha = \frac{4}{5}$$

$$\sin \alpha < 0$$



$$\therefore \tan \alpha = -\frac{3}{4}$$

Question 15



draw a line parallel to the other 2

$x = 46$ alternate angles are equal if the lines are parallel

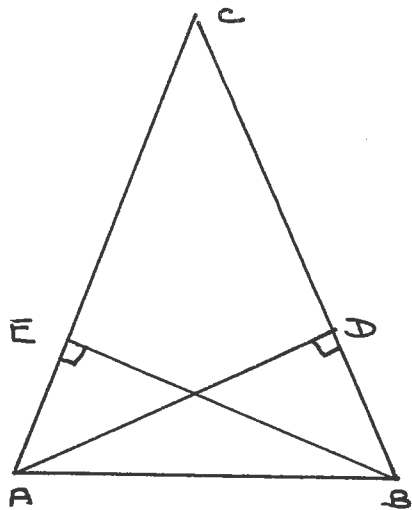
$t = 54$ alternate angles are equal if the lines are parallel

$y = 360 - (46 + 54)$ angles about a point add up to 360°
 $= 260$

$$\begin{aligned}
 \text{b) Interior angle} &= \frac{(n-2) \times 180}{n} \quad n=8 \\
 &= \frac{(8-2) \times 180}{8} \\
 &= 135^\circ
 \end{aligned}$$

$\therefore$ Each interior angle is 135°

c)



In $\triangle ABE$ and $\triangle BAD$

AB is common

$BE = AD$ given

$\hat{AEB} = \hat{ADB}$ given

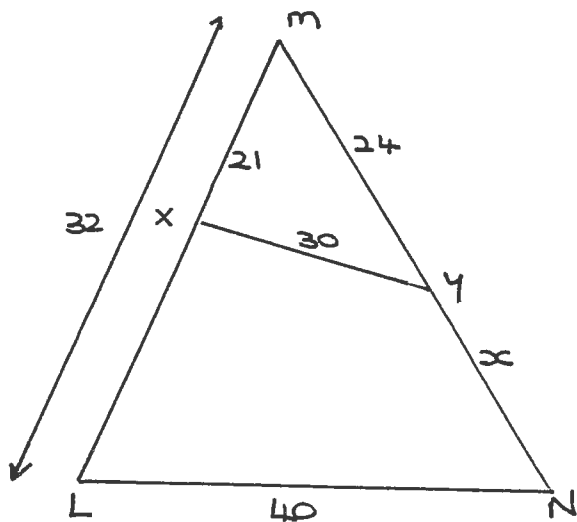
$\therefore \triangle ABE \equiv \triangle BAD$ RHS

$\hat{EAB} = \hat{DBA}$

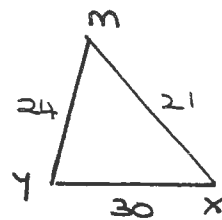
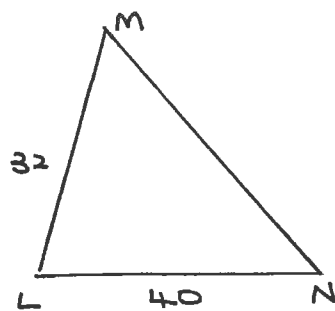
corresponding
angles of congruent
triangles are equal

$\therefore \triangle ABC$ is isosceles
as it has 2 equal
angles

d)



i)



In $\triangle LMN$ and $\triangle MYX$

M is common

$$\frac{LM}{MY} = \frac{32}{24} = \frac{4}{3}$$

$$\frac{LN}{YX} = \frac{40}{30} = \frac{4}{3}$$

$\therefore \triangle LMN \parallel \triangle MYX$

two pairs of corresponding
angles are in proportion
and the included angles
are equal.

$$\begin{aligned} \text{ii) } \frac{LN}{YX} &= \frac{MN}{MY} && \text{corresponding} \\ &&& \text{sides of} \\ \frac{40}{30} &= \frac{24+x}{21} && \text{similar triangles} \\ &&& \text{are in proportion} \end{aligned}$$

$$30(24+x) = 40 \times 21$$

$$24+x = \frac{40 \times 21}{30}$$

$$24+x = 28$$

$$x = 4$$

$$\therefore YN = 4$$

Question 16

a)

$$(i) \quad y = 5x^4 - 7x^2 + 8$$

$$\frac{dy}{dx} = 20x^3 - 14x$$

$$(ii) \quad y = \frac{3x^4 - 5x^2}{x}$$
$$= \frac{x(3x^3 - 5x)}{x}$$

$$= 3x^3 - 5x$$

$$\frac{dy}{dx} = 9x^2 - 5$$

$$(iii) \quad y = \frac{1}{x\sqrt{x}}$$

$$= \frac{1}{x^{3/2}}$$

$$= x^{-3/2}$$

$$\frac{dy}{dx} = -\frac{3}{2} x^{-5/2}$$

$$= \frac{-3}{2\sqrt{x^5}}$$

$$= \frac{-3}{2x^2\sqrt{x}}$$

$$(iv) \quad y = \frac{6}{x^4}$$

$$= 6x^{-4}$$

$$\frac{dy}{dx} = -24x^{-5}$$

$$= \frac{-24}{x^5}$$

$$v) \quad y = (3x+4)^5$$

$$\frac{dy}{dx} = 5(3x+4)^4 \times 3$$

$$= 15(3x+4)^4$$

$$vi) \quad y = x(3-2x)^4$$

$$= u v$$

where

$$u = x$$

$$\frac{du}{dx} = 1$$

$$v = (3-2x)^4$$

$$\frac{dv}{dx} = -8(3-2x)^3$$

$$\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$$

$$= (3-2x)^4 - 8x(3-2x)^3$$

$$= (3-2x)^3 [3-2x-8x]$$

$$= (3-2x)^3 (3-10x)$$

$$vii) \quad y = \frac{x^2+5}{x-2}$$

$$= \frac{u}{v}$$

where

$$u = x^2 + 5$$

$$v = x - 2$$

$$\frac{du}{dx} = 2x$$

$$\frac{dv}{dx} = 1$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$= \frac{2x(x-2) - (x^2+5)}{(x-2)^2}$$

$$= \frac{2x^2 - 4x - x^2 - 5}{(x-2)^2}$$

$$= \frac{x^2 - 4x - 5}{(x-2)^2}$$

$$= \frac{(x-5)(x+1)}{(x-2)^2}$$

$$\text{viii)} \quad y = x \sqrt{1-x^2} \\ = u \quad v$$

$$\text{where } u = x \quad v = (1-x^2)^{\frac{1}{2}} \\ \frac{du}{dx} = 1 \quad \frac{dv}{dx} = \frac{1}{2}(1-x^2)^{-\frac{1}{2}} \times -2x \\ = -x(1-x^2)^{-\frac{1}{2}}$$

$$\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx} \\ = (1-x^2)^{\frac{1}{2}} - x^2(1-x^2)^{-\frac{1}{2}} \\ = (1-x^2)^{-\frac{1}{2}} [1-x^2-x^2] \\ = (1-x^2)^{-\frac{1}{2}} (1-2x^2) \\ = \frac{1-2x^2}{\sqrt{1-x^2}}$$

$$\text{b)} \quad f(x) = x^2 + 4x - 12 \quad f'(x) = -6$$

$$f'(x) = 2x + 4$$

$$\text{But } f'(x) = -6$$

$$\therefore 2x + 4 = -6$$

$$2x = -10$$

$$x = -5$$

when $x = -5$

$$f(x) = x^2 + 4x - 12$$

$$f(-5) = (-5)^2 + 4(-5) - 12$$

$$= 25 - 20 - 12$$

$$= -7$$

$\therefore P$ is the point $(-5, -7)$

$$\text{c)} \quad y = x^2 - 8x + 7$$

$$\frac{dy}{dx} = 2x - 8 \quad \text{at } x = 3$$

$$= 2(3) - 8$$

$$= -2$$

$\therefore$ gradient of tangent is -2

For perpendicular lines

$$m_1 m_2 = -1$$

$\therefore$ gradient of normal is $\frac{1}{2}$

$$y - y_1 = m(x - x_1) \quad (3, -8)$$

$$y + 8 = \frac{1}{2}(x - 3)$$

$$2y + 16 = x - 3$$

$$x - 2y - 19 = 0$$